Unsupervised SAR Image Change Detection for Few Changed Area Based on Histogram Fitting Error Minimization

Kaiyu Zhang, Xiaolei Lv, *Member, IEEE*, Huiming Chai, and Jingchuan Yao

***Abstract*— Change detection in synthetic aperture radar (SAR) images is an essential task of remote sensing image analysis. However, the thresholding procedure is the main difficulty in change detection for a few changed areas for traditional change detection methods. In this article, we propose a novel change detection method for very few changed or even none changed areas. The proposed method contains three procedures: difference image (DI) generation, thresholding, and spatial analysis. In the second procedure, a new thresholding method called histogram fitting error minimization (HFEM) is proposed for a few changed areas. HFEM is derived under the assumption that the unchanged class in the absolute-valued DI follows the half-normal distribution, and the changed class follows the Gaussian distribution. In the spatial analysis procedure, a new conditional random fields (CRF) method based on half-normal distribution is proposed to model the mutual influences among image pixels. The proposed CRF method is called half-normal CRF (HNCRF). Experiments carried out on both synthetic datasets and four real SAR datasets demonstrate the superiority of our method. Not only a few changed datasets but datasets with lots of changes are used in the experiments. The kappa coefficients of the proposed method can reach up to ten times that of the traditional method under extreme conditions. The results prove that the proposed method outperforms the traditional methods in the case of a few changed areas. Meanwhile, the proposed method can get similar results compared with traditional methods under normal conditions.**

***Index Terms*— Change detection, conditional random fields (CRFs), half-normal distribution, image thresholding, synthetic aperture radar (SAR) images, unsupervised change detection.**

Manuscript received 3 January 2022; revised 11 April 2022 and 29 June 2022; accepted 10 July 2022. Date of publication 14 July 2022; date of current version 25 July 2022. This work was supported in part by the LuTan-1 L-Band Spaceborne Bistatic SAR Data Processing Program under Grant E0H2080702 and in part by the China Academy of Railway Sciences Fund under Grant 2019YJ028. *(Corresponding author: Xiaolei Lv.)*

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Digital Object Identifier 10.1109/TGRS.2022.3190977

I. Introduction

CHANGE detection using multitemporal remote sensing images aims to highlight changed regions by analyzing two or more images over the same area acquired at different times, which is one of the most important applications of remote sensing technology [1]–[4]. As a significant application of remote sensing image, change detection analysis provides an effective technological significance for land use and land cover monitoring [5], [6], urban planning and management [7]–[9], natural disaster assessment, monitoring [10]–[12], and so on. Optical remote sensing systems require good sunlight and weather conditions to acquire high-quality optical images. In contrast, remotely sensed synthetic aperture radar (SAR) imagery is adopted widely for change detection, owing to its ability to penetrate cloud cover and its insensitivity to atmospheric and lighting conditions [4], [13], [14].

Depending on the existence of additional ground reference data, change detection techniques can be divided into supervised and unsupervised approaches [15], [16]. Supervised approaches rely on manual labeled data processes, which are labor-intensive and time-consuming [3], [17]. Thus, unsupervised change detection methods have been developed in the literature [1]. As mentioned in [18], the procedure of unsupervised SAR change detection methods involves three steps: 1) difference image (DI) extraction; 2) thresholding; and 3) spatial analysis of the DI. The images should be preprocessed before change detection, such as radiometric correction, image registration [19], [20], and despeckling [21], [22]. The DI extraction procedure mainly contains numerical comparison methods [1], [18], [23] and statistical methods, such as Kullback–Leibler divergence (KLD) [24]. Furthermore, Aiazzi *et al.* [25] proposed a mean shift enforced information theoretic change detection (MS-ITCD) method, which is a robust method for DI extraction. There are two popular numerical comparison methods called change vector analysis (CVA) [18], [26], [27] and log-ratio (LR) change indicator [1], [23]. For single-polarimetric SAR images, CVA is equivalent to the image differencing (ID) method. For the convenience of description, we refer to log-ratio (LR), ID, and other change feature methods, such as KLD and MS-ITCD, collectively as DI. Their relation is shown in Fig. 1.

In this article, we mainly emphasize unsupervised change detection methods, and the LR and ID change indicator is used

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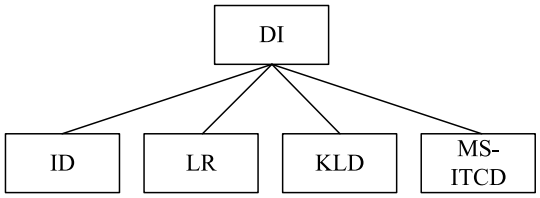


Fig. 1. Relation between DI and different change feature map.

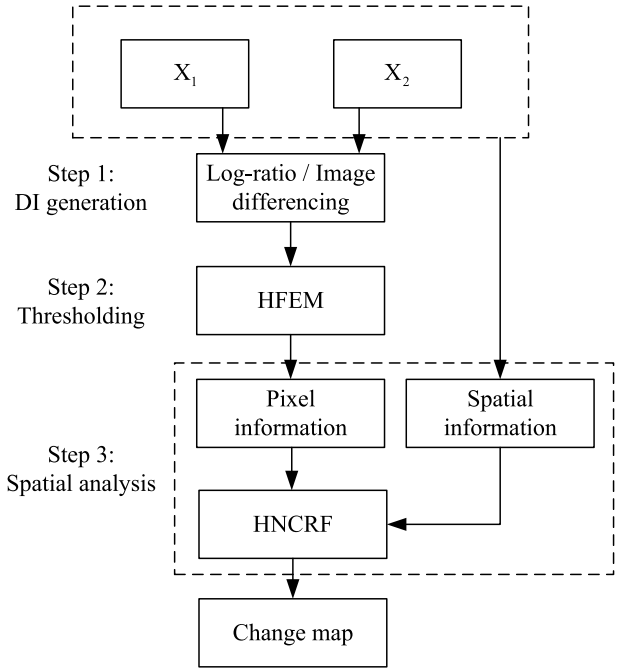


Fig. 2. Framework of the proposed change detection method. The proposed change detection method contains three steps: 1) DI generation using LR or ID indicator. In this article, we use LR for water change detection and ID for building change detection. 2) Thresholding using HFEM. 3) Spatial analysis using HNCRF.

to extract DI. The main contribution of this work lies in the second and third steps.

The adaptive thresholding methods are divided into two types: local thresholding and global adaptive thresholding. Local thresholding techniques compute a threshold value for each pixel neighborhood based on the local statistics and apply it to either the entire neighborhood or only the central pixel [28], [29]. These methods make the standards of different regions inconsistent, which results in many isolated change pixels and holes in the middle of connected change components [28]. Therefore, most studies chose global thresholding method [1], [15], [18], [23], [30], as well as this study. The global thresholding uses all pixels, and the threshold is consistent throughout the image. There are many popular global thresholding methods in change detection, such as Otsu (OTSU) global thresholding method [15], [31], the expectation-maximum (EM) algorithm [18], [32], the K-means algorithm [27], [33], the Kittler–Illingworth (KI) threshold selection algorithm [1], and the fuzzy c-means (FCM) algorithm [30], [34]. However, most of the thresholding algorithms assume that the prior probability of the changed pixels is sufficient to statistically model changed class in the histogram of the difference/ratio image [28]. The basic requirement is that the number of changes should be statistically significant for making the threshold-selection procedure reliable and precise [28]. It means that if there is no change or only a few changes in the area, the existing thresholding method will be imprecise. In [28], they considered splitting a large image into several small blocks and chose appropriate blocks for thresholding. However, the prerequisite for this strategy is that there is a small class of changed pixels in the large-size image. If there is absolutely no change, the split-based approach cannot give robust results. Bazi *et al.* [35] proposed a double-thresholding KI algorithm under the generalized Gaussian (GG) model. When the second-order partial derivatives of the optimization function with respect to the two thresholds are both less than zero, it implies no changes occurred between the two acquisition dates of the two images. Nevertheless, when the changed and unchanged pixels are strongly overlapped, this method may lead to the estimation of biased threshold values. In summary, thresholding for very few changed areas is still an unresolved problem.

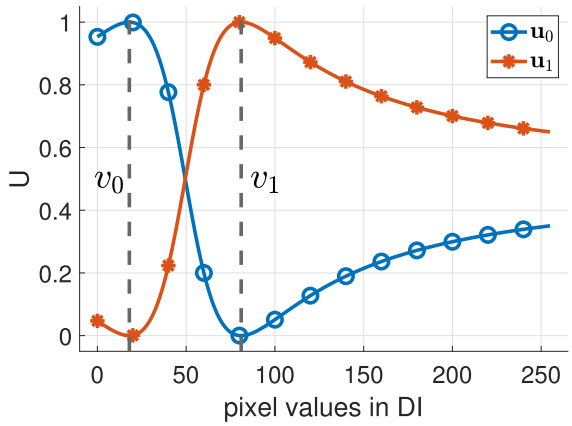


Fig. 3. Curve of $\boldsymbol{u}_0$ and $\boldsymbol{u}_1$. The dependent variables $\boldsymbol{u}_0$ and $\boldsymbol{u}_1$ indicate the degree of membership of the data point in unchanged and changed cluster, respectively. The x-axis denotes the values of pixels in DI.

In order to overcome the aforementioned problems, in this study, we proposed a novel unsupervised thresholding method called histogram fitting error minimization (HFEM). The idea of HFEM contains two points.

1) In the DI, unchanged class follows Gaussian distribution [18] or GG distribution [1], and the expectation of unchanged class should be zeros if two multitemporal images have been radiometrically corrected. However, the unchanged class in the absolute-valued DI does not follow the Gaussian distribution or GG distribution. Therefore, half-normal distribution [36] is introduced to model the statistical distribution of unchanged classes.
2) The traditional thresholding methods require prior knowledge that the image is divided into changed and unchanged classes. If all the pixels belong to the unchanged class, the traditional thresholding methods cannot work well. Thus, we design a novel thresholding method based on half-normal distribution, which does not require this prior knowledge.

The overall performance of the change detection method depends on the quality of all three steps. The third step

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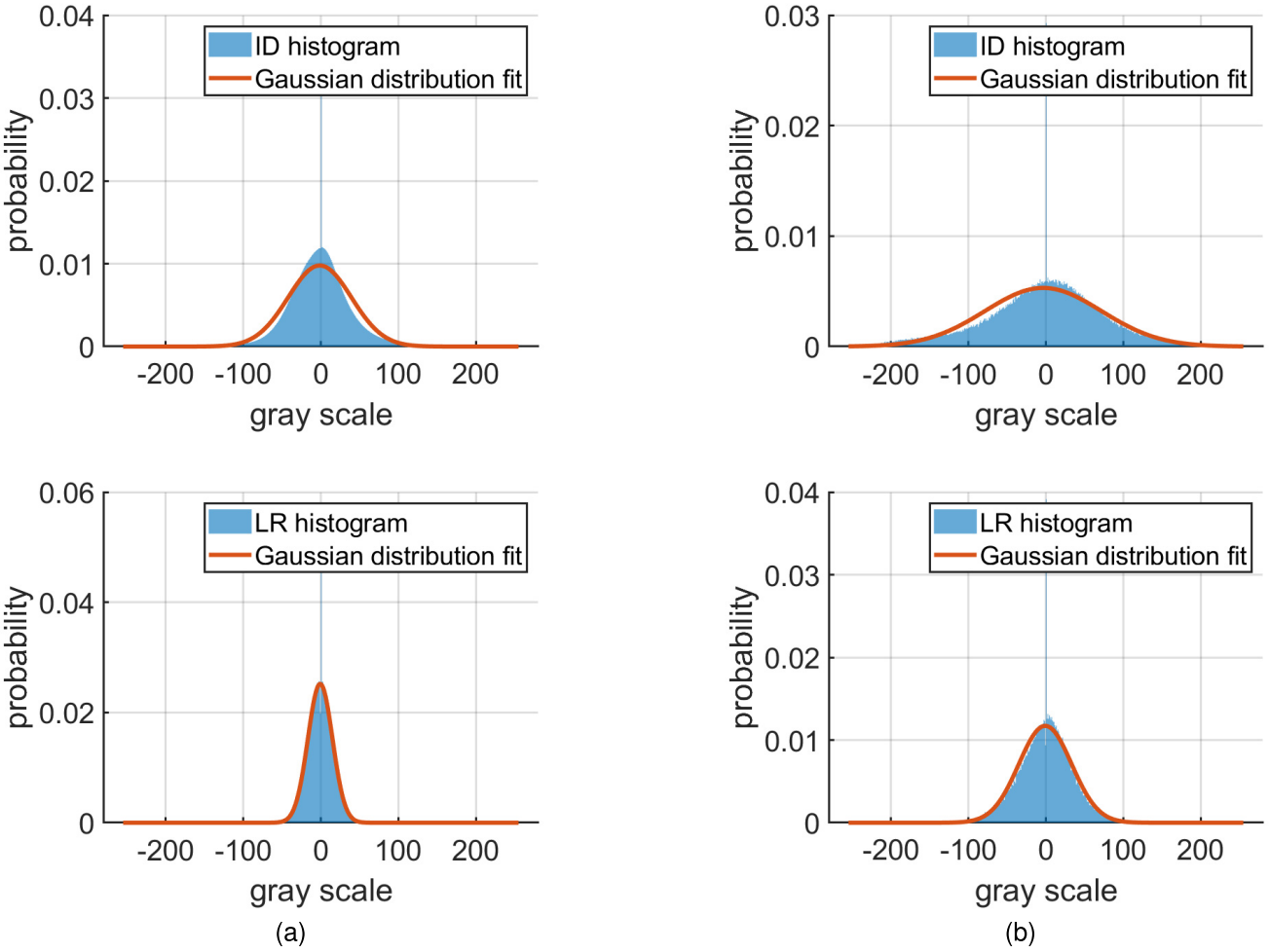


Fig. 4. Histograms of the ID and LR using Tongzhou-large and Tongzhou-small datasets. (a) Tongzhou-large. (b) Tongzhou-small.

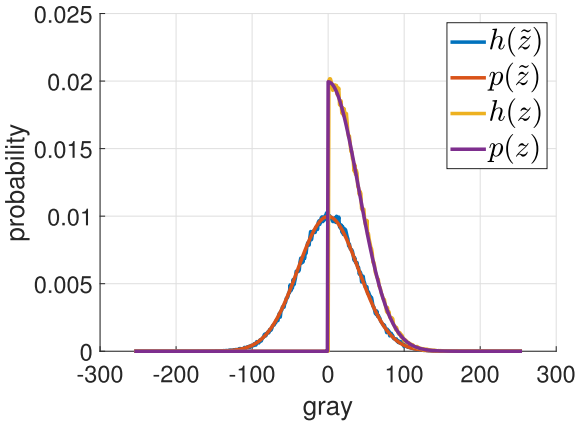


Fig. 5. Schematic of Gaussian distribution and half-normal distribution.

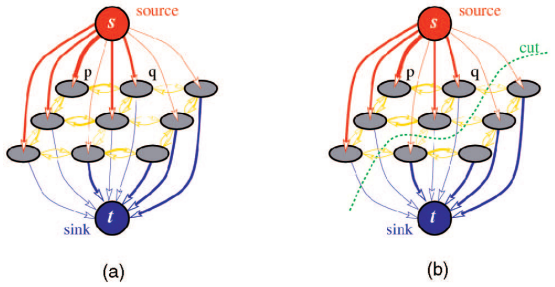


Fig. 6. Example of a directed capacitated graph. (a) Graph G. (b) Cut of G.

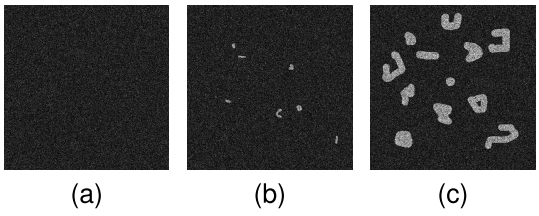


Fig. 7. DI $\boldsymbol{Z}$ of the synthetic data. (a) No change area. (b) Area with a small amount of changes. (c) Area with numerous changes.

of change detection is spatial analysis. In order to model the mutual influences among image pixels, Markov random fields (MRF) have been widely utilized in change detection and image segmentation [18], [30], [37]. Differing from MRF, the conditional random fields (CRFs) is a discriminative probability model which builds the posterior distribution of the label image conditioned on the observed image directly [38], [39]. In this article, CRF is adopted to explore

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TABLE I
VALUES OF THE EVALUATION CRITERIA OF THE SIMULATION

	methods	OA	pre	recall	mIoU	kc
no change	FCM	0.662	0.000	1.000	0.000	0.000
	EM	0.509	0.000	1.000	0.000	0.000
	OTSU	0.668	0.000	1.000	0.000	0.000
	HFEM	**0.999**	0.000	1.000	0.000	0.000
little change	FCM	0.677	0.014	0.980	0.014	0.019
	EM	0.567	0.010	**0.983**	0.011	0.012
	OTSU	0.688	0.015	0.980	0.014	0.020
	HFEM	**0.997**	**0.974**	0.398	**0.394**	**0.564**
much change	FCM	0.952	0.731	0.872	0.660	0.769
	EM	0.917	0.565	**0.920**	0.539	0.655
	OTSU	0.956	0.757	0.862	**0.675**	**0.781**
	HFEM	**0.962**	**0.968**	0.658	0.645	0.764

the spatial information of the image. The CRF model consists of two terms: unary term and pairwise term, which are given and fixed [38]. In this article, we use half-normal distribution to model the unary term of CRF, called half-normal CRF (HNCRF). The experiments demonstrate that HNCRF is superior to Gaussian CRF (GCRF) in change map segmentation.

This article is organized into nine sections. Section II presents the problem formulation and the motivation of the proposed method. Section III describes the fundamental theory and principle of the proposed HFEM and HNCRF. Section IV exploits the proposed novel thresholding method. Section V describes the proposed HNCRF for spatial analysis. The synthetic and real datasets are presented in Section VI, which also contains the experimental design and evaluation criteria. Experimental results on both synthetic and real multitemporal SAR images are reported in Section VII. The discussions of the proposed method are described in Section VIII. Finally, conclusions are drawn in Section IX.

II. PROBLEM FORMULATION AND MOTIVATION

A. Problem Formulation

Let us consider two registered multitemporal SAR intensity images $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$. The radiometric correction has been done on these two images. These two SAR images are acquired in the same area at two different times, T_1 and T_2, respectively. The objective of the change detection framework is to generate the change map that classifies the whole area into two classes: change and no change. The change detection results are displayed in the form of a binary image. In the result, 1 represents a changed pixel and 0 represents an unchanged pixel. The traditional change detection methods mainly contain three steps [18], [23], [30]: DI extraction, thresholding, and spatial analysis. The framework of the proposed method is the same as the traditional methods. Fig. 2 describes the framework of the proposed method.

The LR and the ID are defined as (1)

$$\boldsymbol{Z} = \frac{255}{\log(256)}\left|\log\left(\frac{\boldsymbol{X}_1+1}{\boldsymbol{X}_2+1}\right)\right|$$
$$\boldsymbol{Z} = |\boldsymbol{X}_2 - \boldsymbol{X}_1|. \tag{1}$$

For the convenience of expression, we use $\boldsymbol{Z}$ to represent both ID and LR. To facilitate analysis, the coefficient (255/log(256)) is used to map the range of LR map to $[0, 255]$. According to Bazi *et al.* [1], the LR operator is more powerful than ID in low-intensity regions. Therefore, we use LR for flood change detection because water appears in low-intensity areas in SAR images. When detecting changes in high-intensity regions, such as building change detection, we choose to use ID. The experiment will show the superiority of ID in building change detection. If the "mixed" change is considered, which means an area has both urban and water changes, both LR and ID are used for mixed change detection. First, we calculate the change detection results using LR and ID as DI, respectively. Then, the two change detection results are combined (logical or) to get the final result.

In this article, we choose to focus on the thresholding step and the spatial analysis step. We give a novel thresholding method as well as a novel spatial analysis method. The proposed thresholding method is called HFEM, and the novel spatial analysis method is called HNCRF. The overall framework of our change detection method is depicted in Fig. 2, where HFEM is used in the thresholding step, and HNCRF is used for spatial analysis.

B. Motivation

The motivation of this work is to avoid the problems of traditional thresholding methods. The thresholding procedure is one of the most critical steps in change detection tasks. However, in the case of no change or a few changes in an area, all the existing thresholding methods cannot work. In this section, we take FCM as an example to explain why the current adaptive thresholding methods fail to work.

Let N denote the number of samples and C denote the number of classes. Each sample indicates the DI of a pixel. Let d denote the dimension of the feature of the samples. In the FCM method, $\boldsymbol{U} \in \mathbb{R}^{N\times d}$ is the fuzzy partition matrix. Element $\boldsymbol{U}(i, j)$ indicates the degree of membership of the jth data point in the ith cluster. The matrix $\boldsymbol{U}$ is often used as the prior probability in the change detection field. The objective function of FCM is given as follows [40]:

$$\min_{U,V} J = \sum_{i=1}^{C}\sum_{j=1}^{N} u_{ij}^m \left\|\boldsymbol{z}_j - \boldsymbol{v}_i\right\|^2$$
$$\text{s.t.} \sum_{i=1}^{C} u_{ij} = 1, \quad j = 1, 2, 3, \ldots, N \tag{2}$$

where $\boldsymbol{V} = [v_1, v_2, \ldots, v_C] \in \mathbb{R}^{d\times C}$ represents the matrix composed of k central vectors. $u_{ij} = \boldsymbol{U}(i, j)$. $\boldsymbol{v}_i$ and u_{ij} are calculated using (3) and (4) in each iteration step. The algorithm does not stop until the parameters stop updating in a certain iteration [40]

$$\boldsymbol{v}_i = \frac{\sum_{j=1}^{N} u_{ij}^m \boldsymbol{z}_j}{\sum_{j=1}^{N} u_{ij}^m} \tag{3}$$

$$\boldsymbol{U}(i, j) = \frac{1}{\sum_{k=1}^{C}\left(\frac{\left\|z_j - \boldsymbol{v}_i\right\|}{\left\|z_j - \boldsymbol{v}_k\right\|}\right)^{\frac{2}{m-1}}} \tag{4}$$

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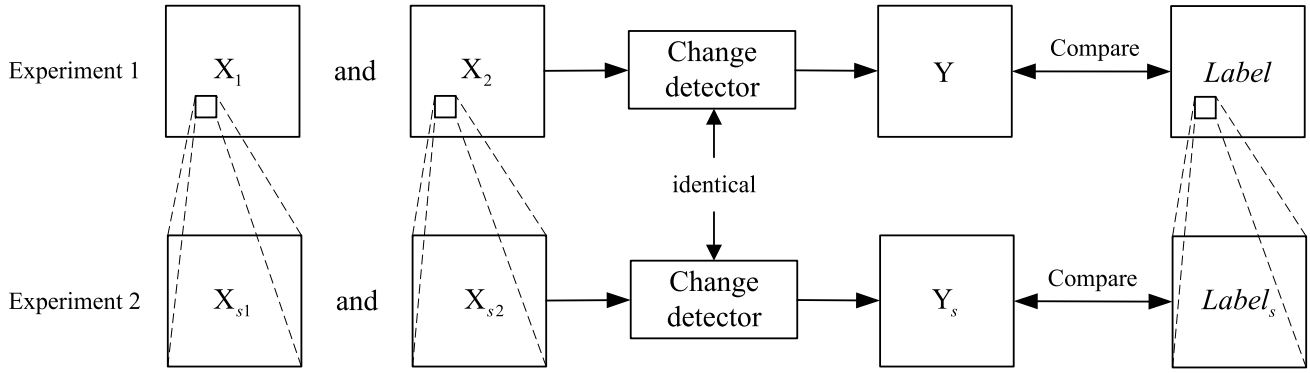


Fig. 8. Schematic of two parts of real data experiment. The first row indicates the experiment using the complete data, and the second row indicates that we use the cropped images from the complete data.

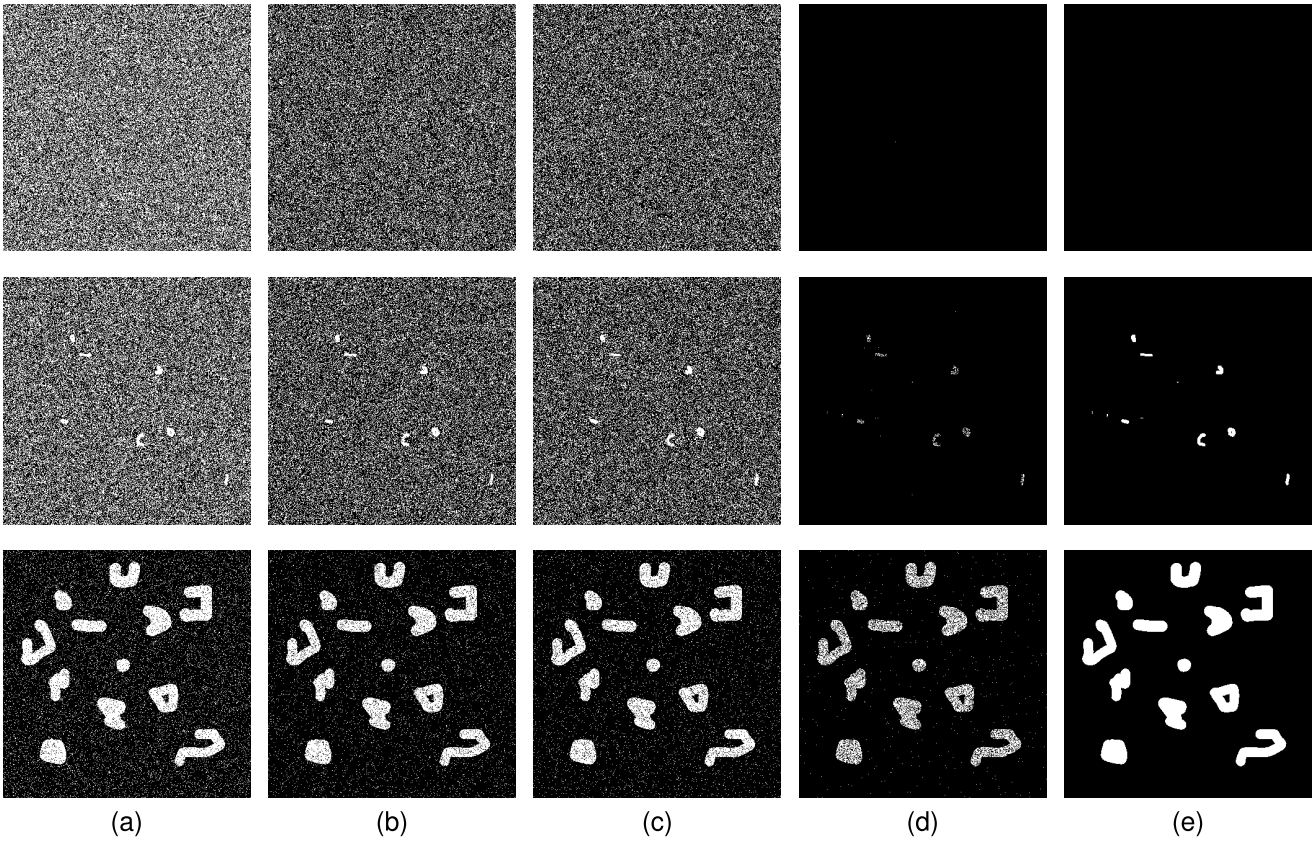


Fig. 9. Simulation test for the proposed thresholding method. The first, second, and third rows illustrate the results of no change, a few change, and many changes. (a) EM. (b) FCM. (c) OTSU. (d) HFEM. (e) Label.

where the vector $\boldsymbol{z}_j \in \mathbb{R}^{d\times 1}$ represents the feature of the jth sample, and the vector $\boldsymbol{v}_i \in \mathbb{R}^{d\times 1}$ represents the center vector of the ith class. The hyperparameter m measures the degree of fuzzy overlap, with $m > 1$. In this article, we set $m = 2$. In change detection applications, $k = 2$, and $\boldsymbol{U} = [\boldsymbol{u}_0, \boldsymbol{u}_1], \boldsymbol{u}_0, \boldsymbol{u}_1 \in \mathbb{R}^{N\times 1}$, where $\boldsymbol{u}_0$ and $\boldsymbol{u}_1$ indicate the degree of membership of the data point in unchanged and changed cluster, respectively. When the number of classes is equal to 2, (4) becomes

$$\boldsymbol{u}_0(j) = \frac{1}{1 + \frac{\|\boldsymbol{z}_j - \boldsymbol{v}_0\|^2}{\|\boldsymbol{z}_j - \boldsymbol{v}_1\|^2}}$$

$$\boldsymbol{u}_1(j) = 1 - \boldsymbol{u}_0(j). \tag{5}$$

In the change detection tasks, $\boldsymbol{z}_j$ often indicates the jth pixel of the DI. If we use the LR map defined in (1), $\boldsymbol{z}_j$ denotes the jth pixel of $\boldsymbol{Z}$. Then, $\boldsymbol{z}_j$ and $\boldsymbol{v}_i$ are both scalars, and (5) becomes two simple unary scalar equations

$$\begin{aligned} \boldsymbol{u}_0(z) &= \frac{1}{1 + \frac{\|z - v_0\|^2}{\|z - v_1\|^2}} \\ \boldsymbol{u}_1(z) &= 1 - \boldsymbol{u}_0(z), \quad z \in [0, 255]. \end{aligned} \tag{6}$$

Before the follow-up discussion, we first prove a lemma.

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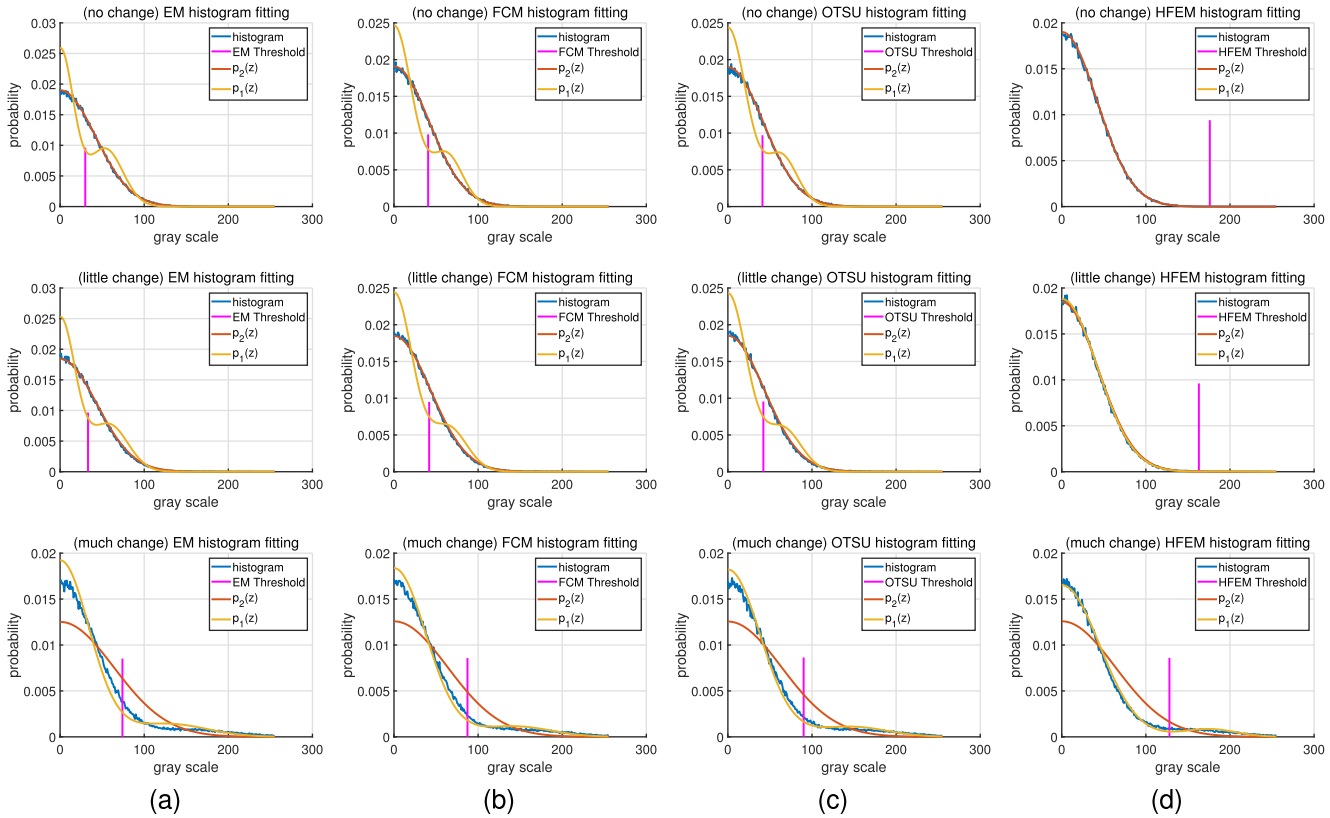


Fig. 10. Histogram fitting for simulation test. The first, second, and third rows illustrate the fitting results of no change, little change, and much change. (a) HFEM. (b) FCM. (c) EM. (d) OTSU.

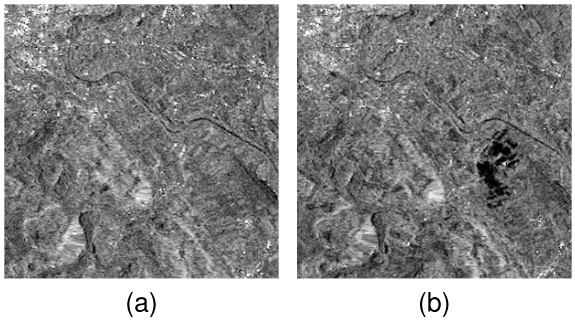


Fig. 11. Bern dataset. (a) $\boldsymbol{X}_1$. (b) $\boldsymbol{X}_2$.

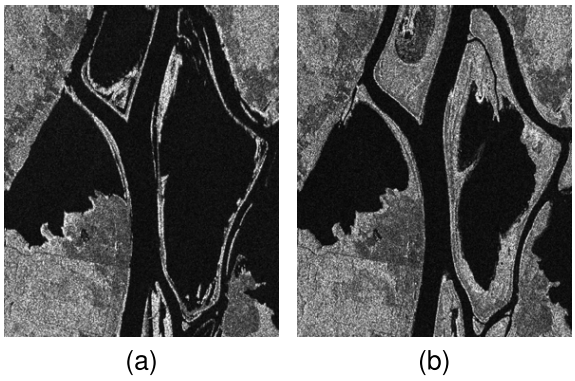


Fig. 12. Ottawa dataset. (a) $\boldsymbol{X}_1$. (b) $\boldsymbol{X}_2$.

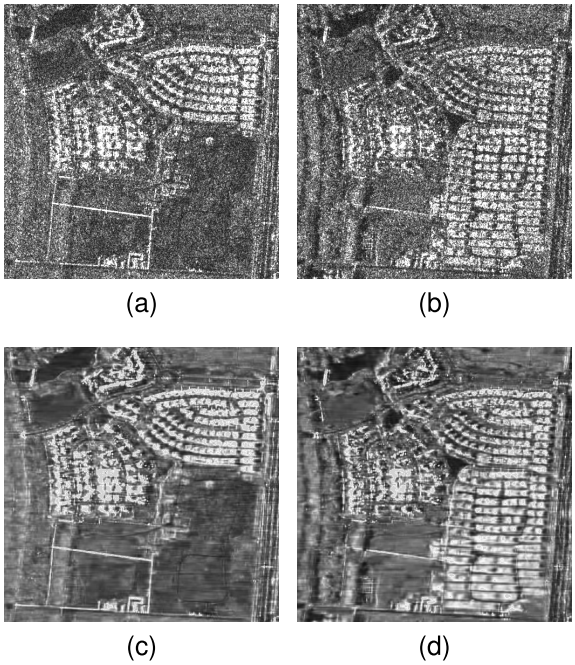


Fig. 13. Tongzhou-small datasets before and after despeckling. The despeckled images are used in the experiments. (a) $\boldsymbol{X}_1$ without despeckling. (b) $\boldsymbol{X}_2$ without despeckling. (c) $\boldsymbol{X}_1$ after despeckling. (d) $\boldsymbol{X}_2$ after despeckling.

Lemma 1: In the FCM method, if z_j and v_i are both scalars, v_0 and v_1 denote the center value of the unchanged and changed classes, respectively. Then, $v_0 > 0$ and $v_1 < z_{\max}$, where $z_{\min}$ and $z_{\max}$ indicate the minimum and the maximum value of $\boldsymbol{Z}$.

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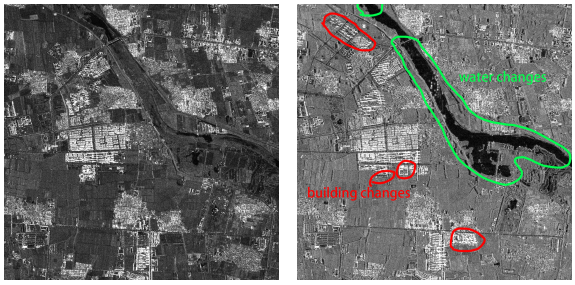


Fig. 14. Tongzhou-mixed-change dataset. (a) $\boldsymbol{X}_1$. (b) $\boldsymbol{X}_2$.

Proof: Assume $v_0 = 0$, then

$$\boldsymbol{u}_0(z) = \frac{1}{1 + \frac{\|z\|^2}{\|z - v_1\|^2}} > 0. \tag{7}$$

According to (3), in the next iteration, v_0 becomes

$$v_0 \leftarrow \frac{\sum_{j=1}^{N} u_{0j}^m z_j}{\sum_{j=1}^{N} u_{0j}^m} > 0. \tag{8}$$

Similarly, assume $v_1 = z_{\max}$, and in the next iteration, v_1 becomes

$$v_1 \leftarrow \frac{\sum_{j=1}^{N} u_{1j}^m z_j}{\sum_{j=1}^{N} u_{1j}^m} < \frac{z_{\max} \sum_{j=1}^{N} u_{1j}^m}{\sum_{j=1}^{N} u_{1j}^m} = z_{\max}. \tag{9}$$

Without loss of generality, we set $v_0 = 18$ and $v_1 = 81$, then we can plot the curve of $\boldsymbol{u}_0$ and $\boldsymbol{u}_1$, as shown in Fig. 3. From Fig. 3, we can see that $\boldsymbol{u}_0$ gets the maximum value at v_0 and the minimum value at v_1. Nevertheless, when $z = 0$, which means the pixel does not change at all, $\boldsymbol{u}_0$ becomes smaller. When $z > v_1$, $\boldsymbol{u}_0$ increases as z increases. In other words, when the difference of the pixel grows larger, the prior probability of this pixel being unchanged gets larger instead. The analysis of $\boldsymbol{u}_1$ can draw the same conclusion. This conclusion seems ridiculous.

Furthermore, the existing thresholding method has two other unreasonable points. First, it can be demonstrated (see Appendix A) that when the SAR images are interfered with by multiplicative speckle noise, the expectation of the unchanged area in the difference map and the LR map is equal to zero. It means v_0 should be equal to zero. However, according to lemma 1, v_0 must be larger than zero. Second, if there is no change in this area, $z_{\max}$ must be small. According to lemma 1, $v_1 < z_{\max}$ and the threshold T must less than v_1. This leads to numerous false alarms.

Similar to the FCM method, the other thresholding method can also lead to similar results. Therefore, the traditional thresholding methods fail to work if the amount of changes is insufficient.

III. Principle of the Proposed Method

In this section, we first analyze the histogram of the DI. Then, we propose a novel half-normal distribution model for change detection. The half-normal distribution can be applied in both the thresholding step and spatial analysis step.

A. Histogram of Log-Ratio Map

In order to analyze the property of the LR map, we generate the LR map of two real datasets, respectively. These two datasets are called the Tongzhou-large dataset and the Tongzhou-small dataset, which are briefly introduced in the following.

The Tongzhou-large dataset was acquired by the TerrSAR-X satellite, which consists of the two SAR images covering the region of Tongzhou, Beijing, China. These two images were obtained in January 2014 and August 2015, respectively. In this article, these two images are performed the multilook operation, and the number of looks in the azimuth direction and the range direction are both 4. The size of the final image is 6900 × 4675, where 6900 is the number of azimuth sampling points. After the multilook operation, the range resolution equals 3.6 m, and the azimuth resolution equals 7.6 m. This dataset reflects the urban development of Tongzhou.

The Tongzhou-small dataset contains two SAR images acquired by TerraSAR-X. These two images are cropped from the Tongzhou-large dataset without multilook operation, which means the range resolution is equivalent to 0.9 m, and the azimuth resolution equals 1.9 m. The size of cropped image is 510 × 510. There were numerous newly built houses in this area between January 2014 and August 2015, so this dataset is used by us for building change detection.

Let $\widetilde{\boldsymbol{Z}}$ denote the ID/LR map without absolute value, which is calculated as follows, respectively:

$$\begin{aligned} \text{ID: } \widetilde{\boldsymbol{Z}} &= \boldsymbol{X}_2 - \boldsymbol{X}_1 \\ \text{LR: } \widetilde{\boldsymbol{Z}} &= \frac{255}{\log(256)} \log\left(\frac{\boldsymbol{X}_1 + 1}{\boldsymbol{X}_2 + 1}\right). \end{aligned} \tag{10}$$

The histograms of $\widetilde{\boldsymbol{Z}}$ of the Tongzhou-large and Tongzhou-small datasets are illustrated in Fig. 4. The first line of Fig. 4 denotes the histograms of ID, and the second line represents the histograms of LR. As we can see from Fig. 4, the histograms of both ID and LR of the two multitemporal SAR images approximately both obey the Gaussian distribution with a mean value of zero. In actual applications, the amount of pixels of the changed area only occupies a small part of the whole image, so it can be considered that the unchanged area follows Gaussian distribution with a mean value of zero. In the meanwhile, there is an obvious impulse at the gray value of zero, which shows that most of the pixels have not changed. This phenomenon further confirms the assumption that the mean value of the unchanged pixels equals zero.

B. Probability Densities of the Difference Image

$\boldsymbol{Z}$ and $\widetilde{\boldsymbol{Z}}$ represent the LR map with and without absolute value. Compared (1) and (10), the relation between $\widetilde{\boldsymbol{Z}}$ and $\boldsymbol{Z}$ is

$$\boldsymbol{Z} = |\widetilde{\boldsymbol{Z}}|. \tag{11}$$

According to the previous analysis, the unchanged pixels in $\widetilde{\boldsymbol{Z}}$ always obey a Gaussian distribution. The question is which distribution of the unchanged pixels in $\boldsymbol{Z}$ obey.

Let $p(\widetilde{z}|\omega_u)$ and $p(z|\omega_u)$ denote the PDF of unchanged pixels in $\widetilde{\boldsymbol{Z}}$ and $\boldsymbol{Z}$, respectively. Let ω_u and ω_c represent the prior probabilities for changed and unchanged classes.

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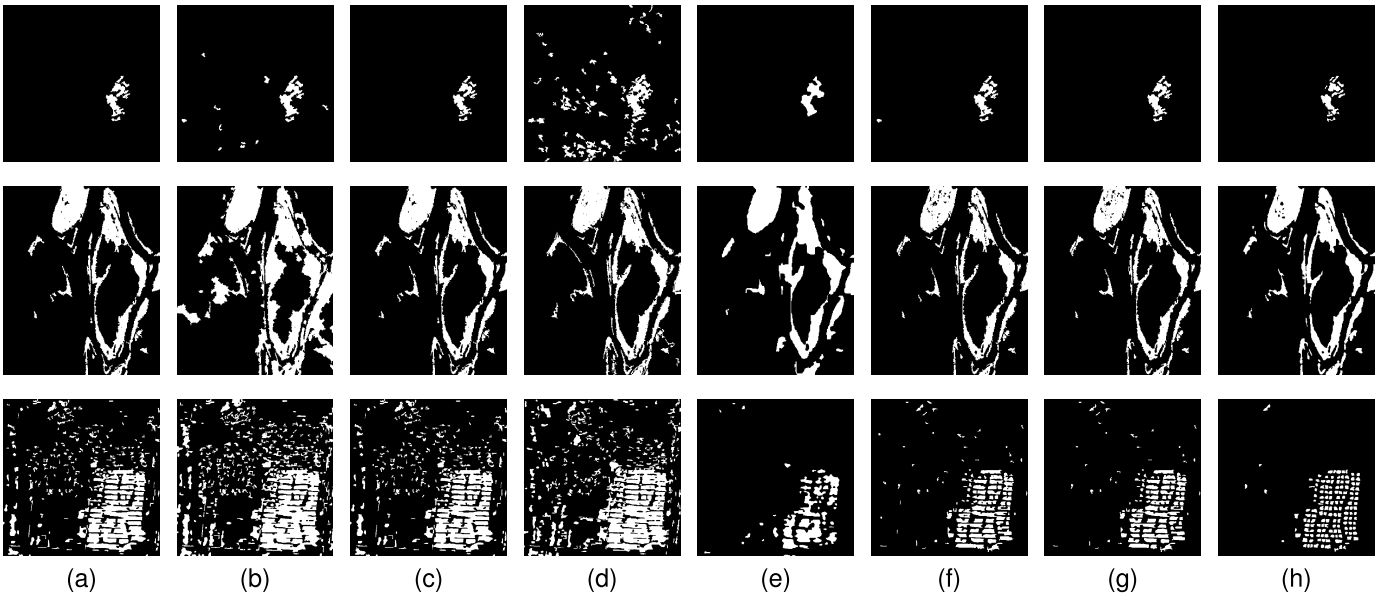


Fig. 15. Results of experiment 1 using Bern, Ottawa, and Tongzhou complete datasets. The first, second, and third rows illustrate the results of Bern, Ottawa, and Tongzhou datasets, respectively. (a)–(g) Change detection results of different methods, as shown in the second column of Table III. (h) Label.

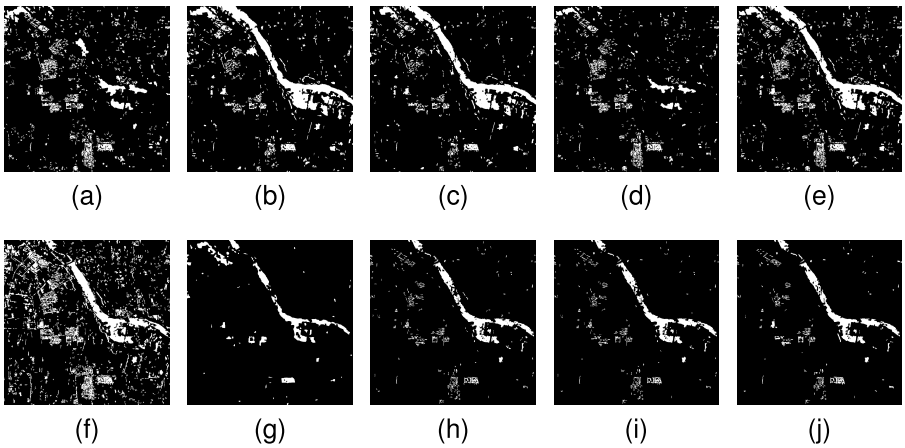


Fig. 16. Results of experiment 1 using Tongzhou mixed-CD complete datasets. (a)–(i) Change detection results of different methods, as shown in the second column of Table III. (j) Label.

In $\widetilde{\boldsymbol{Z}}$, these pixel values obey Gaussian distribution and the expectation of the unchanged area in $\widetilde{\boldsymbol{Z}}$ is equal to zero according to Appendix A. Therefore, $p(\widetilde{z}|\omega_u)$ is defined as follows:

$$p\left(\widetilde{z}|\omega_u\right) = \frac{1}{\sqrt{2\pi}\,\sigma_u} e^{-\frac{\widetilde{z}^2}{2\sigma_u^2}}. \tag{12}$$

To calculate $p(z|\omega_u)$, we first calculate the cumulative distribution function (CDF) for the unchanged pixels in $\widetilde{\boldsymbol{Z}}$

$$\begin{aligned} P(\widetilde{\boldsymbol{Z}} \le z) &= \int_{-\infty}^{z} p(t|\omega_u)dt \\ &= \frac{1}{2} + \frac{1}{\sqrt{2\pi}\,\sigma_u}\int_0^z \exp\left(-\frac{t^2}{2\sigma_u^2}\right)dt. \end{aligned} \tag{13}$$

Then, calculate the CDF for the unchanged pixels in $\boldsymbol{Z}$. Just change the lower limits of the definite integral, and we can get

$$\begin{aligned} P(\boldsymbol{Z} \le z) &= \int_{-z}^{z} p(t|\omega_u)dt \\ &= 2\frac{1}{\sqrt{2\pi}\,\sigma_u}\int_0^z \exp\left(-\frac{t^2}{2\sigma_u^2}\right)dt \\ &= 2\left(P(\widetilde{\boldsymbol{Z}} \le z) - \frac{1}{2}\right). \end{aligned} \tag{14}$$

Taking the derivative of (14), we can get the PDF of unchanged pixels in $\boldsymbol{Z}$

$$p(z|\omega_u) = \frac{2}{\sqrt{2\pi}\,\sigma_u} e^{-\frac{z^2}{2\sigma_u^2}}, \quad z \ge 0 \tag{15}$$

where σ_u denotes the standard deviation of the unchanged pixels.

Equation (15) is called a half-normal distribution, which was originally introduced by Bland and Altman [36]. In order to facilitate understanding, we randomly generate a Gaussian random image $\widetilde{\boldsymbol{Z}}$ with a mean value of 0 and a standard deviation of 40. We, respectively, fit $\widetilde{\boldsymbol{Z}}$ and $\boldsymbol{Z}$ with the Gaussian distribution and the half-normal distribution. The histograms and fitting results are shown in Fig. 5.

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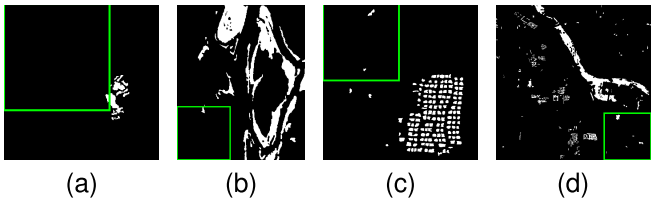


Fig. 17. Cropped area in the labels. (a) Bern dataset. (b) Ottawa dataset. (c) Tongzhou-small dataset. (d) Tongzhou-mixed-change dataset. The green box represents the boundary of the cropped area.

TABLE II
COMPARISON OF ID AND LR USING TONGZHOU DATASET. HFEM AND HNCRF ARE USED IN STEP 2 AND STEP 3, RESPECTIVELY

	OA	pre	recall	mIOU	kc
ID	**0.968**	**0.669**	**0.930**	**0.637**	**0.761**
LR	0.942	0.508	0.898	0.480	0.620

In Fig. 5, $h(\widetilde{z})$ and $h(z)$ denote the histogram of $\widetilde{\boldsymbol{Z}}$ and $\boldsymbol{Z}$, respectively. $p(\widetilde{z})$ and $p(z)$ denote the fit probability density function of corresponding data.

We assume that the changed pixels in $\boldsymbol{Z}$ obey the Gaussian distribution. Let σ_c represent the standard deviation of the changed pixels and μ_c denote the expectation of the changed pixels. The posterior probability $p(z|\omega_c)$ of the changed pixels is defined as the Gaussian distribution

$$p(z|\omega_c) = \frac{1}{\sqrt{2\pi}\sigma_c} e^{-\frac{(z-\mu_c)^2}{2\sigma_c^2}}. \tag{16}$$

Finally, we can get the overall probability of $\boldsymbol{Z}$. The overall probability can be divided into two cases. The first case is that there are some changes in this area, and the second case is that there is absolutely no change. Therefore, let $p_1(z)$ and $p_2(z)$ denote the overall PDF in these two cases, respectively. If there is a change in the area, according to Bayes theorem, $p_1(z)$ could be written as follows:

$$p_1(z) = p(z|\omega_u)P(\omega_u) + p(z|\omega_c)P(\omega_c) \tag{17}$$

where $z \in [0, 255]$. $P(\omega_u)$ and $P(\omega_c)$ represent the prior probability of the unchanged class and the changed class, respectively.

If there is no change in the multitemporal images, the overall probability of $\boldsymbol{Z}$ obeys half-normal distribution, and the expression of $p_2(z)$ becomes

$$p_2(z) = \frac{2}{\sqrt{2\pi}\sigma} e^{-\frac{z^2}{2\sigma^2}}, \quad z \in [0, 255] \tag{18}$$

where σ represents the standard deviation of all samples.

In Section IV, our method is mainly developed based on the four equations of formula (15) to formula (18).

IV. HISTOGRAM FITTING ERROR MINIMIZATION

In this section, we present a novel thresholding method based on the probability densities of the DI. The novel thresholding method is called HFEM.

A. Objective Function

The most important thing about the thresholding method is the objective function. OTSU thresholding method is determined by minimizing intraclass intensity variance, or equivalently, by maximizing interclass variance [31]. The EM algorithm is an iterative method to find (local) maximum likelihood or maximum *a posteriori* (MAP) estimates of parameters in statistical models [32]. The FCM aims to minimize an objective function defined as (2).

The objective function of the proposed thresholding method is to minimize the histogram fitting error

$$\min_z \sum_{z=0}^{255} (p_1(z) - h(z))^2 \tag{19}$$

where $h(z)$ represents the histogram of $\boldsymbol{Z}$.

B. Constraints

In addition, in order to make our segmentation algorithm cope with both many and a few changes, two constraints need to be added.

1) Maximum Posterior Probability Criterion: According to the maximum posterior probability criterion, we have

$$\hat{\omega} = \arg\max_{\omega=\{\omega_u,\omega_c\}} p(\omega|z). \tag{20}$$

It is obvious that the larger z, the greater the probability of change. Then, we have

$$\begin{cases} p(\omega_u|z) \le p(\omega_c|z), & z \le T \\ p(\omega_u|z) > p(\omega_c|z), & z > T. \end{cases} \tag{21}$$

If the gray level $z = T$, then

$$p(\omega_u|z) = p(\omega_c|z). \tag{22}$$

According to the Bayes theorem

$$p(\omega|z) = \frac{P(\omega)p(z|\omega)}{p(z)}.$$

Equation (22) can be written as

$$P(\omega_u)p(z|\omega_u) = P(\omega_c)p(z|\omega_c). \tag{23}$$

2) Better Fitting Criterion: According to the previous discussions, If there is indeed a change in the multitemporal images, the fitting effect of $p_1(z)$ should be better than $p_2(z)$. Let $h(z)$ represent the histogram of $\boldsymbol{Z}$. The better fit criterion is defined as inequality (24)

$$\sum_{z=0}^{255} (p_1(z) - h(z))^2 < \sum_{z=0}^{255} (p_2(z) - h(z))^2. \tag{24}$$

C. Model Solving

In summary, our objective function is to minimize the fitting error under the above-mentioned two conditions. Therefore, the optimization model of the algorithm for solving the threshold T is defined as follows:

$$\begin{aligned} T = \arg\min_z & \sum_{z=0}^{255} (p_1(z) - h(z))^2 \\ \text{s.t.} & \sum_{z=0}^{255} (p_1(z) - h(z))^2 < \sum_{z=0}^{255} (p_2(z) - h(z))^2 \\ & |P(\omega_u)p(z|\omega_u) - P(\omega_c)p(z|\omega_c)| < \text{eps} \end{aligned} \tag{25}$$

where eps represents a positive number with small absolute value.

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To solve the optimization model (25), we only need to traverse z between [1, 255]. Why the traversal process does not start from zero is that it will make $P(\omega_u)$ equal to zero, making the denominator be zero, as shown in the fourth equation in (26). The input of the algorithm is the histogram $h(z)$ of $\boldsymbol{Z}$. According to (15) to (18), we can calculate the parameters of the PDF using $h(z)$

$$\begin{aligned} P(\omega_u) &= \sum_{z=0}^{T} h(z) \\ P(\omega_c) &= \sum_{z=T+1}^{255} h(z) \\ \sigma^2 &= \sum_{z=0}^{255} (z-0)^2 h(z) \\ \sigma_u^2 &= \frac{1}{P(\omega_u)} \sum_{z=0}^{T} (z-0)^2 h(z) \\ \mu_c &= \frac{1}{P(\omega_c)} \sum_{z=T+1}^{255} h(z) z \\ \sigma_c^2 &= \frac{1}{P(\omega_c)} \sum_{z=T+1}^{255} (z-\mu_c)^2 h(z). \end{aligned} \tag{26}$$

It is worth noting that both $p_2(z)$ and $p(z|\omega_u)$ obey the half-normal distribution, and the parameter estimation is different from those of the Gaussian distribution. When calculating the variance of $p_2(z)$ and $p(z|\omega_u)$, one should calculate according to the third and the fourth formulas in (26) instead of using sample variance.

The thresholding algorithm process is summarized as Algorithm (1). The proposed thresholding method aims to minimize the fitting error of the histogram of DI, so we call the proposed thresholding method HFEM.

Algorithm 1 Thresholding Algorithm of DI

Input: The histogram $h(z)$ and the number of pixels N;
Output: Optimal threshold T;
1: Calculate σ^2 using the third formula in (26);
2: Calculate $p_2(z)$ using formula (18);
3: Let the optimal threshold candidate set be $\Omega_T = \emptyset$
4: **for** $z = 1 \rightarrow 255$ **do**
5: Update parameters using (26);
6: Calculate $p(z|\omega_u)$, $p(z|\omega_c)$ and $p_1(z)$ according to Eq. (15)to Eq. (18);
7: **if** T meets the constraints of (25) **then**
8: $\Omega_T = \Omega_T \bigcup z$
9: **end if**
10: **end for**
11: Find the value that minimizes the objective function of the (25) in the set Ω_T;
12: **return** T

V. Proposed Half-Normal Conditional Random Field

In this section, we first review the classical MRF and CRF, and then the CRF model is modified under the condition that the unchanged area follows half-normal distribution, named HNCRF. HNCRF is used to combine neighborhood information to make up for the lack of pixel-level change detection.

A. Markov Random Fields

Let $\boldsymbol{X} = \text{concat}(\boldsymbol{X}_1, \boldsymbol{X}_2), \boldsymbol{X} \in \mathbb{R}^{H \times W \times 2}$ represent the two-channel image after two input images are concatenated. $\boldsymbol{x}_i, \boldsymbol{x}_j$ represent two pixels in $\boldsymbol{X}$, and $\mathcal{N}^8$ denotes eight neighborhoods, $(i, j) \in \mathcal{N}^8$.

Let us represent the two-channel image as a set of N pixels, $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]$. Let $f_i \in \{\omega_u, \omega_c\}$ indicate the labels that each pixel may be assigned to. The main objective is to assign an appropriate label to each pixel in the image $\boldsymbol{X}$.

Let $P(f) = P(\boldsymbol{X} = f)$ $f = \{f_1, f_2, \ldots, f_N\}$ represent the overall probability of the image assigned the label set f. If $\boldsymbol{X}$ is called a MRF, it should satisfy the Markov property [41]

$$P\left(f_i | f_{\boldsymbol{X}-\boldsymbol{x}_i}\right) = P\left(f_i | f_{\mathcal{N}_i}\right). \tag{27}$$

The formula (27) means the probability of pixel x_i assigned to f_i is only related to the label of the neighboring pixels. According to the Hammersley–Clifford theorem, the MRF is equivalent to the Gibbs distribution [39]

$$P(f) = \frac{1}{Z} \exp\left\{ \sum_{c \in C} U(f_c) \right\} \tag{28}$$

$$Z = \sum_{f} \exp\left\{ \sum_{c \in C} U(f_c) \right\} \tag{29}$$

where Z denotes the normalization factor. The set C represents the set of all possible neighborhood relation combinations. If we use $\mathcal{N}_i$, then the formula (28) can be written as

$$P(f) = \frac{1}{Z} \exp\left\{ \sum_{i=1}^{N} \sum_{j \in \mathcal{N}_i} \psi\left(f_i, f_j\right) \right\} \tag{30}$$

where $\psi(f_i, f_j)$ denotes the pairwise function between label f_i and f_j.

B. Conditional Random Fields

Different from MRF, Lafferty *et al.* [38] directly consider the conditional probability $P(f|\boldsymbol{X})$. The assumption of the CRFs is that f is defined as the MRF under a given condition $\boldsymbol{X}$. In this case, f is called CRFs. The posterior probability of CRF can be written as follows:

$$P(f|\boldsymbol{X}) = \frac{1}{Z(\boldsymbol{X})} \exp\left\{ \sum_{i=1}^{N} \phi(\boldsymbol{x}_i, f_i) + \sum_{i=1}^{N} \sum_{j \in \mathcal{N}_i} \psi\left(f_i, f_j, \boldsymbol{x}_i, \boldsymbol{x}_j\right) \right\} \tag{31}$$

where $Z(X)$ is the normalization factor, $\phi(\boldsymbol{x}_i, f_i)$ denotes the unary function, and $\psi(f_i, f_j, \boldsymbol{x}_i, \boldsymbol{x}_j)$ represents the pairwise relation.

Regarding unsupervised learning, the form of the unary function is generally proportional to the logarithm of $P(\boldsymbol{x}_i|f_i)$ [2], [39]. The form of the pairwise function

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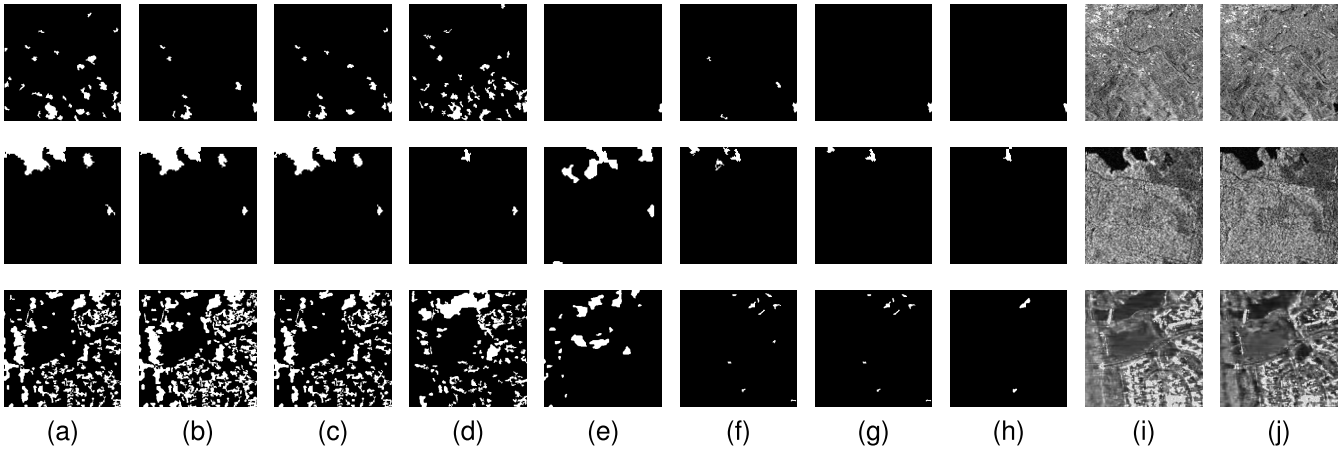


Fig. 18. Results of experiment 2 using Bern, Ottawa, and Tongzhou cropped datasets. The first, second, and third rows illustrate the results of Bern, Ottawa, and Tongzhou datasets, respectively. For Bern, Ottawa, and Tongzhou datasets, the percentages of change with respect to the number of pixels of the analyzed patch are 0.20%, 0.42%, and 0.40%, respectively. (a)–(g) Change detection results of different methods, as shown in the second column of Table IV. (h) Label. (i) $\boldsymbol{X}_1$. (j) $\boldsymbol{X}_2$.

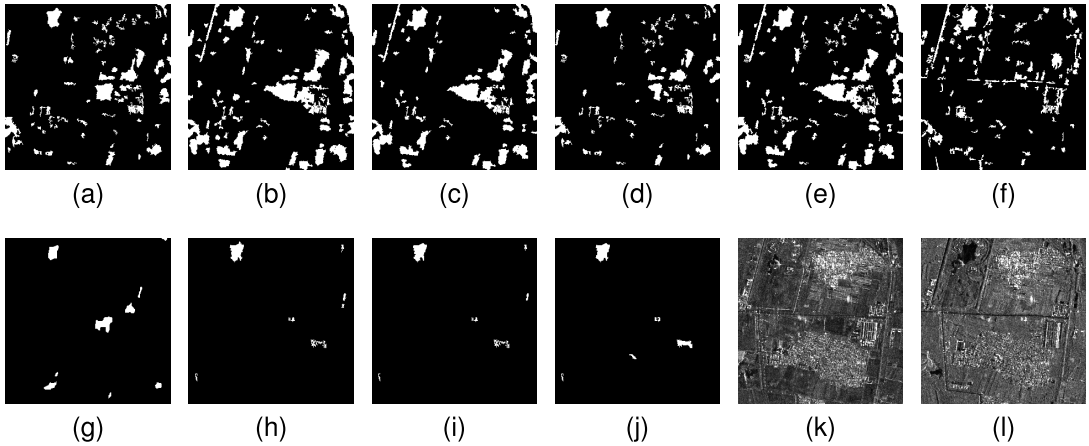


Fig. 19. Results of experiment 2 using the cropped Tongzhou mixed-CD complete datasets. For Tongzhou-mixed-change dataset, the percentage of change with respect to the number of pixels of the analyzed patch is 1.08%. (a)–(i) Change detection results of different methods, as shown in the second column of Table III. (j) Label. (k) $\boldsymbol{X}_1$. (l) $\boldsymbol{X}_2$.

$\psi(f_i, f_j, \boldsymbol{x}_i, \boldsymbol{x}_j)$ representing the pairwise relation needs to be set as appropriate according to the different situations.

The goal of MRF and CRF is to obtain the largest posterior probability $P(f|\boldsymbol{X})$ [42].

C. Half-Normal Conditional Random Fields

In change detection tasks, the unary function depends on the DI, so the unary function in 31 becomes

$$\phi(\boldsymbol{x}_i, f_i) = \phi(z_i, f_i) \tag{32}$$

where z_i represent the ith pixel of the DI $\boldsymbol{Z}$.

According to the specific form of the unary function, we define HNCRF.

Definition 1: CRFs defined by (31) become half-normal conditional random fields if the two following conditions are satisfied.

1) $p(z|f_i = \omega_u)$ is modeled by Half-Normal distribution defined by (15).
2) $p(z|f_i = \omega_c)$ is modeled by the Gaussian distribution defined by (16).
3) $\phi(z_i, f_i)$ is proportional to the logarithm of $p(z_i|f_i)$.

Bruzzone and Prieto [18] assume that both $p(z_i|f_i = \omega_u)$ and $p(z_i|f_i = \omega_c)$ can be modeled by Gaussian distributions. Gong *et al.* [30] consider a multivariate GG model for these two probabilities. Lv *et al.* [39] choose a membership value fuzzy partition matrix $\boldsymbol{U}$ in FCM as the cost of pixel $p(z_i|f_i)$. In this article, we use HNCRF to model the unary function

$$\begin{cases} \phi(z_i, \omega_u) = 2\log[p(z_i|\omega_u)] \\ \qquad\qquad = -\dfrac{z_i^2}{\sigma_u^2} - \log\left(2\pi\sigma_u^2\right) + 2\log(2) \\ \phi(z_i, \omega_c) = 2\log[p(z_i|\omega_c)] \\ \qquad\qquad = -\dfrac{(z_i - \mu_c)^2}{\sigma_c^2} - \log\left(2\pi\sigma_c^2\right). \end{cases} \tag{33}$$

For the sake of simplicity, we write the pairwise function $\psi(f_i, f_j, \boldsymbol{x}_i, \boldsymbol{x}_j)$ as ψ_{ij}. The pairwise function is the same as

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TABLE III
COMPARISON OF DIFFERENT METHODS USING THE COMPLETE DATASETS

	method index	DI generation	thresholding	spatial analysis	OA	pre	recall	mIoU	kc
Bern	(a)	LR	FCM	HNCRF	**0.996**	0.816	0.916	0.759	0.861
	(b)	LR	EM	HNCRF	0.990	0.571	**0.939**	0.551	0.706
	(c)	LR	OTSU	HNCRF	0.997	**0.828**	0.913	**0.767**	**0.866**
	(d)	MS-ITCD	OTSU	HNCRF	0.947	0.190	0.981	0.190	0.304
	(e)	GKLD	OTSU	HNCRF	0.993	0.699	0.797	0.594	0.742
	(f)	LR	HFEM	GCRF	0.995	0.750	0.932	0.711	0.829
	(g)(Ours)	LR	HFEM	HNCRF	**0.996**	0.788	0.929	0.742	0.850
Ottawa	(a)	LR	FCM	HNCRF	**0.975**	0.916	0.905	**0.853**	**0.906**
	(b)	LR	EM	HNCRF	0.885	0.582	**0.968**	0.571	0.660
	(c)	LR	OTSU	HNCRF	0.975	0.935	0.906	0.853	0.906
	(d)	MS-ITCD	OTSU	HNCRF	0.964	0.855	0.929	0.802	0.869
	(e)	GKLD	HFEM	HNCRF	0.922	0.749	0.764	0.609	0.711
	(f)	LR	HFEM	GCRF	0.973	0.959	0.863	0.832	0.892
	(g)(Ours)	LR	HFEM	HNCRF	0.973	**0.966**	0.857	0.831	0.892
Tongzhou-small	(a)	ID	FCM	HNCRF	0.884	0.340	**0.989**	0.339	0.458
	(b)	ID	EM	HNCRF	0.815	0.245	0.994	0.245	0.328
	(c)	ID	OTSU	HNCRF	0.884	0.340	**0.989**	0.339	0.458
	(d)	MS-ITCD	OTSU	HNCRF	0.832	0.262	0.986	0.261	0.353
	(e)	GKLD	FCM	HNCRF	0.945	0.541	0.597	0.396	0.538
	(f)	ID	HFEM	GCRF	0.965	0.646	0.942	0.621	0.749
	(g)(Ours)	ID	HFEM	HNCRF	**0.968**	**0.669**	0.930	**0.637**	**0.761**
Tongzhou-mixed-change	(a)	ID	FCM	HNCRF	0.927	0.355	0.638	0.295	0.420
	(b)	LR	FCM	HNCRF	0.930	0.406	0.965	0.400	0.541
	(c)	LR	OTSU	HNCRF	0.947	0.474	0.956	0.464	0.609
	(d)	ID	OTSU	HNCRF	0.940	0.397	0.492	0.282	0.408
	(e)	LR + ID	OTSU	HNCRF	0.925	0.390	**0.979**	0.387	0.525
	(f)	MS-ITCD	OTSU	HNCRF	0.882	0.286	0.968	0.283	0.397
	(g)	GKLD	OTSU	HNCRF	0.967	0.687	0.559	0.446	0.599
	(h)	LR + ID	HFEM	GCRF	0.992	0.911	0.924	0.848	0.914
	(i)(Ours)	LR + ID	HFEM	HNCRF	**0.995**	**0.969**	0.919	**0.893**	**0.941**

previous works [9], [30], [43]

$$\begin{cases} \psi_{ij} = \exp\left(-\dfrac{\left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|^2}{2\sigma^2}\right) \cdot \dfrac{1}{d(i,j)} \\ \sigma^2 = \left\langle \left\|\boldsymbol{x}_i - \boldsymbol{x}_j\right\|^2 \right\rangle \end{cases} \tag{34}$$

where $\langle\cdot\rangle$ represents the average operator of all pairs of neighboring pixels. $d(i,j) \in \{1, \sqrt{2}\}$ is the spatial distance of the $\mathcal{N}^8$ neighboring pixels.

Considering the balanced parameter λ, which balances the weight of the unary term and the pairwise term, the final posterior probability of HNCRF is defined as follows:

$$P(f|\boldsymbol{X}) = \frac{1}{Z(\boldsymbol{X})} \exp\left\{\lambda \sum_{i=1}^{N} \phi(z_i, f_i) + \sum_{i=1}^{N} \sum_{j \in \mathcal{N}_i} \psi_{ij}\right\}. \tag{35}$$

We use the globally optimized graph-cut algorithm [44] as the solution algorithm. The algorithm regards the image $\boldsymbol{X}$ as a graph, where each pixel represents a node. At the same time, a source node s and a target node t are connected with the other nodes. The constructed graph is shown in Fig. 6, and the picture is from [44]:

The source node s represents the foreground, which means the changed area. In contrast, the target node t represents the background. There are two connection ways in the graph. The first connection way is called t-link, which corresponds to the red edge and blue edge in Fig. 6(a). Since s represents the foreground, we use the unary function $\phi_i(z_i, \omega_c)$ as the weight of the red edge. Similarly, t represents the background, so we use $\phi_i(z_i, \omega_u)$ as the weight of the blue edge. The second connection way called n-link is the connection between pixels, which corresponds to the yellow part in the graph (6)(a). According to the algorithm described in [44], it should be

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TABLE IV
COMPARISON OF DIFFERENT METHODS USING THE CROPPED DATASETS

	method index	DI generation	thresholding	spatial analysis	OA	pre	recall	mIOU	kc
Bern	(a)	LR	FCM	HNCRF	0.952	0.040	0.988	0.040	0.073
	(b)	LR	EM	HNCRF	0.989	0.151	0.977	0.161	0.275
	(c)	LR	OTSU	HNCRF	0.979	0.089	0.988	0.089	0.160
	(d)	MS-ITCD	OTSU	HNCRF	0.939	0.032	**1.000**	0.032	0.352
	(e)	GKLD	HFEM	HNCRF	0.999	0.823	0.651	0.571	0.727
	(f)	LR	HFEM	GCRF	0.997	0.393	0.965	0.388	0.558
	(g)(Ours)	LR	HFEM	HNCRF	**0.999**	**0.856**	0.965	**0.830**	**0.907**
Ottawa	(a)	LR	FCM	HNCRF	0.931	0.058	**1.000**	0.058	0.103
	(b)	LR	EM	HNCRF	0.936	0.062	**1.000**	0.062	0.110
	(c)	LR	OTSU	HNCRF	0.934	0.060	**1.000**	0.060	0.106
	(d)	MS-ITCD	HFEM	HNCRF	0.996	0.558	0.869	0.515	0.678
	(e)	GKLD	EM	HNCRF	0.945	0.061	0.836	0.061	0.107
	(f)	LR	HFEM	GCRF	0.990	0.279	0.820	0.263	0.412
	(g)(Ours)	LR	HFEM	HNCRF	**0.996**	**0.515**	0.820	**0.463**	**0.631**
Tongzhou-small	(a)	ID	FCM	HNCRF	0.796	0.019	**0.996**	0.019	0.030
	(b)	ID	EM	HNCRF	0.708	0.013	**0.996**	0.013	0.019
	(c)	ID	OTSU	HNCRF	0.802	0.020	**0.996**	0.020	0.031
	(d)	MS-ITCD	OTSU	HNCRF	0.818	0.020	0.940	0.020	0.032
	(e)	GKLD	OTSU	HNCRF	0.940	0.033	0.494	0.032	0.055
	(f)	ID	HFEM	GCRF	**0.994**	0.361	0.671	0.308	0.467
	(g)(Ours)	ID	HFEM	HNCRF	**0.994**	**0.394**	0.627	**0.319**	**0.481**
Tongzhou-mixed-change	(a)	ID	FCM	HNCRF	0.921	0.116	0.947	0.115	0.190
	(b)	LR	FCM	HNCRF	0.856	0.066	0.939	0.066	0.106
	(c)	LR	OTSU	HNCRF	0.878	0.077	0.930	0.077	0.125
	(d)	ID	OTSU	HNCRF	0.937	0.140	0.926	0.138	0.229
	(e)	LR + ID	OTSU	HNCRF	0.860	0.069	**0.953**	0.069	0.111
	(f)	MS-ITCD	OTSU	HNCRF	0.915	0.106	0.926	0.106	0.175
	(g)	GKLD	OTSU	HNCRF	0.981	0.270	0.427	0.198	0.322
	(h)	LR + ID	HFEM	GCRF	0.997	0.859	0.822	0.724	0.838
	(i)(Ours)	LR + ID	HFEM	HNCRF	**0.997**	**0.904**	0.797	**0.735**	**0.845**

given a higher weight for similar pixels. The ψ_{ij} function meets this requirement, so ψ_{ij} is used as the weight of n-link edges. The graph-cut algorithm is to find the minimum cut of graph G and divide graph G into foreground and background. The minimum cut is shown by the green dashed line in Fig. 6(b).

D. Postprocessing

In the postprocessing step, small fragments in the change map are removed to improve the accuracy. A fragment whose area is less than P pixels is removed. The fragment removal step is also implemented for all the compared methods in the experiments.

VI. DATASETS DESCRIPTIONS AND EXPERIMENTAL SETTINGS

In the following, we first give the evaluation criteria and dataset descriptions used in the experiment. Then, the experimental design is explained in detail. As for the two hyperparameters in the spatial analysis step, we set $\lambda = 0.31$ and $P = 25$ in the experiments, which is a great balance between the unary term and the pairwise term. We give discussions for these two hyperparameters in Section VIII.

A. Evaluation Criteria

The evaluation criteria are based on the confusion matrix. The four elements in confusion matrix are TP, FP, TN, and FN. TP is the number of pixels of the intersection of the detected foreground and the label foreground, and TN is the number of pixels of the intersection of the background and the real background in the detection output. In the ground truth map, the total amount of pixels that really change is recorded as N_c, and the total amount of pixels that do not change is denoted as N_u, then FP and FN are defined as follows:

$$\mathrm{FP} = N_u - \mathrm{TN}$$

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$$\mathrm{FN} = N_c - \mathrm{TP}. \tag{36}$$

The evaluation criteria in this article include overall accuracy (OA), precision, recall, mIOU, and kappa coefficient (KC), which are defined as follows:

$$\begin{aligned} \mathrm{OA} &= \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \\ \mathrm{mIOU} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \\ \mathrm{precision} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \mathrm{recall} &= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}. \end{aligned} \tag{37}$$

KC is defined as follows:

$$\mathrm{kc} = \frac{\mathrm{OA}-\mathrm{PE}}{1-\mathrm{PE}} \tag{38}$$

where

$$\mathrm{PE} = \frac{(\mathrm{TP}+\mathrm{FP})\cdot N_c + (\mathrm{FN}+\mathrm{TN})\cdot N_u}{N^2}. \tag{39}$$

B. Synthetic Dataset for Simulation

According to the previous discussion, the DI without absolute value $\widetilde{\boldsymbol{Z}}$ approximately follows the Gaussian distribution with a mean value of 0. Based on this conclusion, we first generate an image following Gaussian distribution with the size of 400 × 400, which has a mean value of 0 and a standard deviation of 30. The background is regarded as an unchanged area. Then, we select some pixels as changed area. The changed pixels also follow the Gaussian distribution. The mean value of the changed class is set to 150, and the standard deviation is set to 45. We generated three sets of simulation data, no change, a small number of changes, and numerous changes, respectively. For a few changed images, the number of changed pixels only accounts for 0.48% of the total pixels, which is far lower than the assumption of 10% of changed pixels in [28]. For numerous changes, the number of changed pixels accounts for 10.54% of the total pixels. Finally, take the absolute value of $\widetilde{\boldsymbol{Z}}$ to obtain $\boldsymbol{Z}$, as shown in Fig. 7.

C. Real SAR Datasets

We use four real SAR datasets in the experiment. The real datasets include the Bern dataset, the Ottawa dataset, and the Tongzhou-mixed-change dataset, as well as the Tongzhou-small dataset described in Section III. Tongzhou-small images are very high resolution (VHR) images so that the speckle noise in the image is more serious. Therefore, we use the multitemporal SAR block-matching 3-D (MSAR-BM3D) method [22] for despeckling before change detection tasks. Now, we briefly introduce the Bern dataset, the Ottawa dataset, and the Tongzhou-mixed-change dataset.

The Bern dataset includes two 301 × 301 SAR images, which were acquired by the European Remote Sensing Satellite 2 (ERS 2) SAR satellite covering Bern city. The two images were acquired in April and May 1999. Between these two moments, the waters of the Aare River flooded parts of the cities of Thun and Bern, including the Bern Airport.

The Ottawa dataset was acquired by the RADARSAT SAR sensor in the Ottawa region. Two images (290 × 350) were acquired in May and August 1997, respectively. During this time, the area suffered from flooding.

The Tongzhou-mixed-change dataset was cropped from the Tongzhou-large dataset. Two images of the Tongzhou large dataset were obtained in January 2014 and August 2015, respectively. The size of two images of the Tongzhou-mixed-change dataset is 1000 × 1000. Both water changes and building changes took place between T_1 and T_2. In this article, we only care about the overall change without distinguishing the semantic information of the change.

D. Experimental Design

In this work, we implement both simulation and real data experiments to demonstrate the effectiveness of our method. The real data experiment includes two parts. The real datasets used in the first part contain a large number of changes, which is statistically significant for thresholding. It aims to prove that the proposed method is not weaker than other methods in the presence of a large area of change. The second part is the core part of the experiment. We crop the raw images and labels and perform the experiment using cropped data. The number of changes in the cropped area is very small. Experiments prove that our method is far superior to other algorithms in this extreme case.

The way of real data experiments is shown in Fig. 8. Experiment 1 is carried out using complete datasets, and experiment 2 is carried out using cropped datasets.

VII. Experiment

A. Simulation Test

In the simulation test, we use simulation data to verify the effectiveness of HFEM in Section IV. The simulation tests are performed in three situations, where there is no change, a small number of changes, and a large number of changes. The simulation results demonstrate that HFEM performs well in all three situations, while the other methods cannot.

We use three thresholding algorithms as the compared methods, called EM, FCM, and OTSU. The output result of the simulation test is shown in Fig. 9, and the histogram fitting result is shown in Fig. 10. The numerical results are shown in Table I. It can be seen that all methods fit well in the case of a lot of changes. However, the EM, FCM, and OTSU algorithms perform poorly with a few changes and no changes. It means that these methods cannot automatically judge whether there are changes in the area, while HFEM can adaptively determine whether there is a change.

B. Experiment 1: Complete Datasets

Experiment 1 aims to prove that the proposed method can achieve similar results compared with other methods when the dataset distribution is great (more pixels change). The Bern dataset, Ottawa dataset, Tongzhou-small dataset, and Tongzhou-mixed-change dataset are shown in Figs. 11–14, respectively. The two original Tongzhou SAR images are illustrated in Fig. 13(a) and (b), and the despeckled images are illustrated in Fig. 13(c) and (d). The proposed method uses HFEM for thresholding and HNCRF for spatial analysis. First, we tested different images, including LR, ID, MS-ITCD, and KLD. The first step of MS-ITCD is Gaussian smoothing. Following [25], the size of the Gaussian smoothing window is set

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to 7×7, for best performance. KLD uses the 9×9 window to estimate local histograms for best performance. Second, in order to assess the effectiveness of HFEM, we replace HFEM with other threshold methods, including EM, FCM, and OTSU. Finally, we use HNCRF in the spatial analysis step. In order to demonstrate the effectiveness of HNCRF, we use the Gaussian distribution suggested in [18] to model the unary function in (35). We call this compared algorithm GCRF. GCRF is used to assess the superiority of HNCRF.

The result of the Bern dataset is shown in the first row of Fig. 15, the result of the Ottawa dataset is shown in the second row of Fig. 15, and the result of the Tongzhou dataset is shown in the third row of Fig. 15. From Table III, it can be seen that the KC of our method is 1.7% lower than the best index using the Bern dataset. Under the Ottawa dataset, the KC of our method is lower by 1.3% compared with the best index. The visible result of the Tongzhou mixed-CD dataset is shown in Fig. 16. Under the Tongzhou dataset, our method is the most superior. This proves that the proposed method can achieve satisfactory results when there are a large number of changes.

In experiment 1, the datasets contain a large number of changed pixels. It is statistically significant to model the changed and unchanged class, so the traditional methods work well. In this case, the proposed method also obtains a good change detection performance because the maximum posterior probability criterion is used as one of the constraints of the thresholding model.

In this experiment, the ID method is used for building change detection and the LR method is used for water change detection. For water change detection, the LR method has been demonstrated to be more effective than the ID method according to the previous research [1], [30], [45]. However, building areas tend to have higher gray values, so the ID method is better than LR for building change detection. Table II illustrates the comparison between the ID and LR methods using the Tongzhou-small dataset. HFEM and HNCRF are used in the step 2 and step 3 of the change detection framework, respectively. It can be seen that the ID method is superior to the LR method on the Tongzhou-small building change detection dataset.

Furthermore, KLD and MS-ITCD are also famous DI generation methods. Both KLD and MS-ITCD consider the local area to obtain DI, while ID and LR only consider the pixel level comparison. The feature space of KLD and MS-ITCD is not linear for thresholding. Therefore, thresholding methods, such as OTSU, FCM, and HFEM, are not suitable for KLD and MS-ITCD. The final binary change maps are influenced by the thresholding method, so the numerical results may differ even a lot.

C. Experiment 2: Cropped Datasets

The main superiority of the proposed method is embodied in the very few changed areas, where there are a few changed objects or even no changes at all. Simulation tests demonstrate that HFEM can be used for thresholding for areas with no changes and very few changes. In this experiment, we only consider regions with very few changes on the existing datasets. This experiment demonstrates that when the data distribution is not ideal, the superiority of the proposed method far exceeds that of the other algorithms. As shown in Fig. 17, the base maps are the labels of the three datasets, where the green box represents the boundary of the cropped area. For the Bern, Ottawa, Tongzhou, and Tongzhou-mixed-change datasets, the percentages of change with respect to the number of pixels of the analyzed patch are 0.20%, 0.42%, 0.40%, and 1.08%, respectively.

From Figs. 18 and 19 and Table IV, we can see that if there are very few changes in the cropped area, all the other thresholding methods do not work well, while the proposed method can acquire better results than other algorithms.

On the Bern dataset, the KC of HFEM is more than ten times that of the FCM, more than three times that of the EM algorithm, and more than five times that of the OTSU algorithm; in the cropped region of the Ottawa dataset, the KC of HFEM is more than six times that of the FCM algorithm; in the cropped region of the Tongzhou dataset, the KC is more than ten times that of all other algorithms. All the experimental results prove that HFEM is far stronger than other thresholding methods in this case. The traditional thresholding methods cannot be robust in the case of regions with very few changed pixels. Different from traditional methods, HFEM considers the overall probability distribution instead of the posterior probability distribution. It compares the fitting error of $p_1(z)$ and $p_2(z)$. In the case of no change or a few changes, the histogram fitting error of $p_2(z)$ is smaller than that of $p_1(z)$. This makes the optimal threshold keep rising until the optimization model meets the constraints.

As for HNCRF, it is a little weaker than GCRF when using the Bern datasets, but HNCRF is much superior to GCRF in the Ottawa and Tongzhou datasets. Experiment 2 demonstrates that HFEM far exceeds the other thresholding methods, and HNCRF is superior to GCRF. It also demonstrates the rationality of half-normal distribution in DI analysis.

VIII. Discussion

In this section, we first discuss how parameter settings affect accuracy. Then, we quantitatively evaluate the relation between the change detection performance using HFEM and the number of change samples. Finally, we discuss the impact of speckle on the proposed method.

A. Analysis of Parameter Settings

There are two hyperparameters in the proposed method, called λ and P. We discuss the impact of these two parameters on the final detection accuracy. We use both complete and cropped datasets to test the effect of these two parameters.

λ represents the weight factor of the unary term in the CRF model. The physical meaning of λ is that the larger the value of λ, the closer the final result is to the pixel thresholding result of HFEM. Meanwhile, the smaller the value of lambda, the more obvious the effect of neighborhood information. P represents the largest small fragment to be removed in the postprocessing step. KC and IoU are tested in this analysis.

The relations between these two parameters and the performance are shown in Figs. 20 and 21, respectively. The first row of Figs. 20 and 21 illustrates the relation using the

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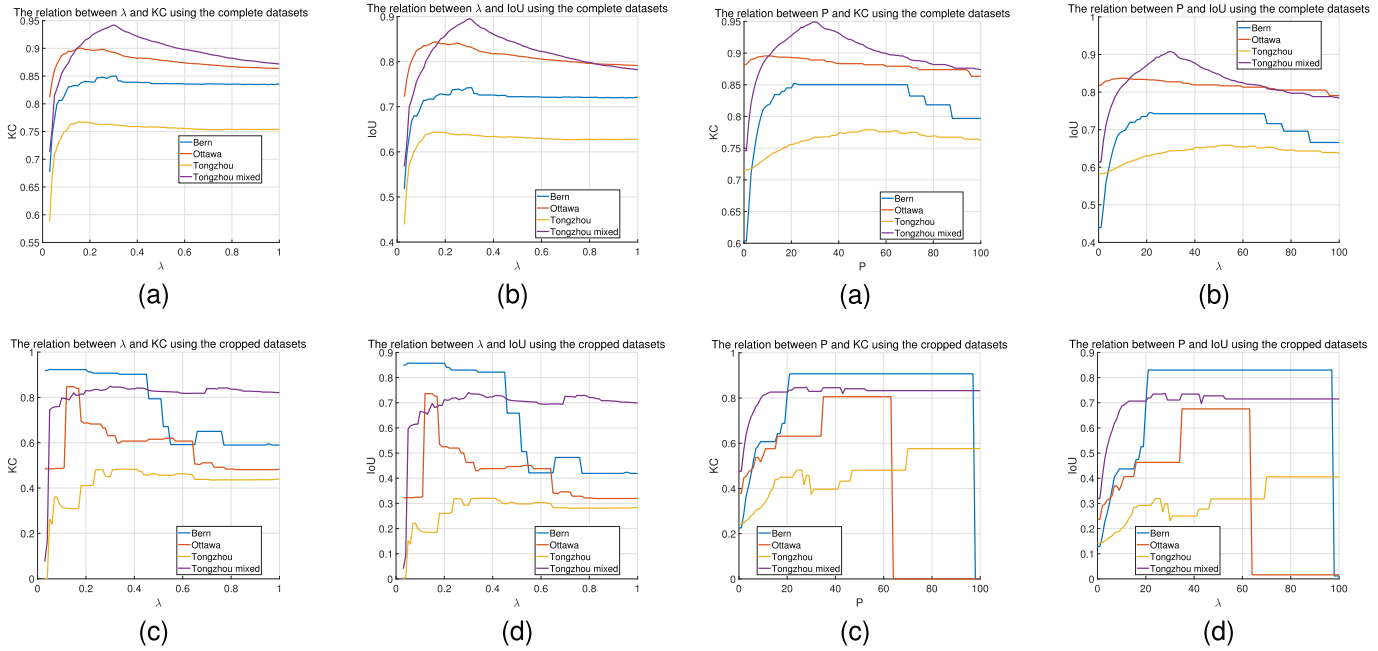


Fig. 20. Relation between λ and the detection performances. (a) λ and KC using the complete datasets. (b) λ and IoU using the complete datasets. (c) λ and KC using the cropped datasets. (d) λ and IoU using the cropped datasets.

Fig. 21. Relation between P and the detection performances. (a) P and KC using the complete datasets. (b) P and IoU using the complete datasets. (c) P and KC using the cropped datasets. (d) P and IoU using the cropped datasets.

complete datasets, and the second row shows the relation using the cropped datasets.

From Fig. 20, we can see that the detection performance increases first and then decreases as λ becomes larger. If the λ is too small, the detection performance will drop sharply. The reason is that when $\lambda \to 0$, (35) becomes

$$P(f|\boldsymbol{X}) = \frac{1}{Z(\boldsymbol{X})} \exp\left\{\sum_{i=1}^{N} \sum_{j \in \mathcal{N}_i} \psi_{ij}\right\}. \tag{40}$$

The graph cut algorithm aims to maximize $P(f|\boldsymbol{X})$. According to the definition of ψ_{ij} in (34), if $\boldsymbol{x}_i = \boldsymbol{x}_j$ for all pixels, then $P(f|\boldsymbol{X})$ defined in (40) takes the maximum value.

The meaning of P is obvious. The main function of P is to remove noise fragments. If P is too large, the detected plate will be removed. On the contrary, if P is too small, some noise fragments cannot be removed. According to Figs. 20 and 21, $\lambda = 0.3$ and $P = 25$ are a good compromise.

B. Performance of HFEM With the Number of Changed Samples

In this section, we randomly generate 0% to 20% changed pixels to evaluate the performance of HFEM. As in the aforementioned simulation tests, the mean value and the standard deviation changed pixels are set to 150 and 45; the mean value and the standard deviation unchanged pixels are set to 0 and 30. The simulation test results are depicted in Fig. 22.

From Fig. 22, we can see that HFEM is far stronger than other methods when the changed pixel ratio is less than 5%. However, when the number of changes should be statistically significant, HFEM is a little inferior to OTSU and FCM.

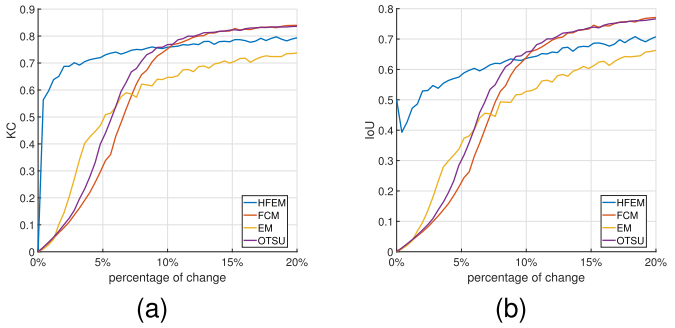


Fig. 22. Simulation test results. The performance of HFEM with the number of changed samples. The horizontal axis represents the proportion of changed pixels, and the vertical axis represents the change detection performance. (a) KC. (b) mIoU.

C. Analysis of the Impact of Speckle Noise

In this article, the despeckling step should be done before change detection. For single-look images, we need to use despeckling before subsequent processing; for multilook images, we do not need to use despeckling because multilook images have less speckle noise.

Fig. 23 illustrates the thresholding results for the Tongzhou dataset with and without despeckling. LR is used to generate DI. For some images with serious speckle, if the speckle noise is not removed, the change of pixel value due to noise is even larger than the real change. For other thresholding algorithms (EM, FCM, and OTSU), if there is too much noise, there will be many false alarms, while the HFEM algorithm is the opposite. If there is too much speckle noise, HFEM will also count the real changes as noise, and there is almost no change in the final segmentation result.

Finally, let us discuss how the despeckling step affects the probability distribution and the histogram fitting effect. HFEM

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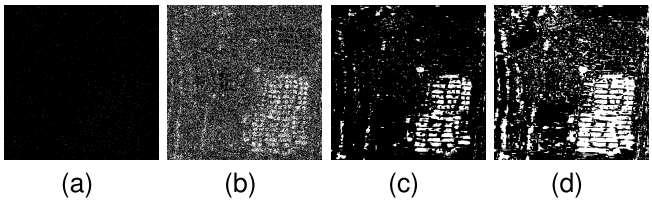


Fig. 23. Thresholding results for Tongzhou original dataset and Tongzhou despeckled dataset. (a) HFEM for original dataset. (b) OTSU for original dataset. (c) HFEM for despeckled dataset. (d) OTSU for despeckled dataset.

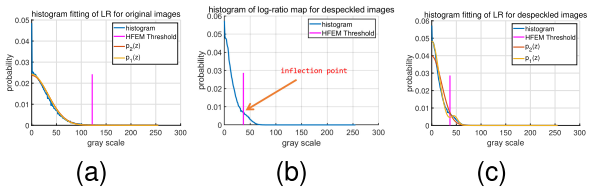


Fig. 24. Thresholding results for Tongzhou original dataset and Tongzhou despeckled dataset. (a) Histogram fitting and HFEM threshold of LR for original images. (b) Histogram of LR for despeckled images. (c) Histogram fitting and HFEM threshold of LR for despeckled images.

is based on the histogram. The quality of the thresholding effect mainly depends on whether the changed and unchanged classes have obvious boundaries on the histogram. For a heavily speckle corrupted image, the histogram of the LR map is shown in Fig. 24:

Intuitively, it can be seen from Fig. 24(a) that for the images with serious noise pollution, the histogram of the LR map appears to be a relatively smooth curve. There is no obvious inflection point in the curve. It can also be seen from Fig. 24(a) that $p_1(z)$ and $p_2(z)$ almost coincide. Therefore, HFEM will treat the whole image as a class. This leads to most parts of the image being unchanged area. That is why HFEM can adaptively segment the rarely changed area or even the unchanged area without many false alarms.

For the despeckled image, the histogram of the LR map is shown in Fig. 24(b). It can be clearly seen from Fig. 24(b) that the histogram curve has an obvious inflection point at the threshold. At this time, HFEM can adaptively calculate a more accurate threshold and distinguish the histograms on both sides of the inflection point.

IX. Conclusion

In this article, a novel change detection method is proposed for very few changed or none changed areas. We first take FCM as an example to analyze why traditional thresholding methods do not provide reliable results. Then, we introduce half-normal distribution to model the unchanged classes of the absolute value of the LR map. Both the mathematical analysis and real datasets demonstrate the rationality of the assumption. Based on half-normal distribution, we propose a novel threshold method as well as the modified CRF. The proposed method can adaptively decide whether a change occurred between the two acquisition dates of the two images, even if the amount of change is not statistically significant. If there is no change at all, the proposed method can also provide reliable results. For larger amounts of changes, some traditional thresholding methods, such as FCM and OTSU, do perform better than HFEM. But HFEM is not too far from state-of-the-art methods when LR and ID are used as change features. As for KLD and MS-ITCD, due to the nonlinearity of the change feature space, the final binary change maps are influenced by the thresholding method. In the future, some novel change detection frameworks can be explored based on nonlinear change features, such as KLD and MS-ITCD.

This thresholding method is derived under the assumption that the unchanged class in the absolute value of the DI follows half-normal distribution, and the unchanged class follows Gaussian distribution. The proposed method is a constrained optimization problem whose objective function is to minimize the fitting error of the histogram. Thus, we call the proposed thresholding method HFEM. The optimization model of HFEM contains two constraints. The first constraint is the well-known maximum posterior probability criterion, and the second constraint is the proposed better fitting criterion. In order to improve the robustness of the proposed method, we use CRF to model the mutual influences among image pixels. The half-normal distribution is also used to model the unary term of CRF, which is called HNCRF.

The effectiveness of the proposed method is tested using both the synthetic and real SAR datasets. On the one hand, the proposed method is not weaker than other methods in the presence of a large area of change. On the other hand, when the regions contain very few changes, the superiority of the proposed method far exceeds that of other algorithms.

It is worth noting that the proposed method is based on the half-normal distribution for the absolute value of the LR map. However, the LR map without GG distribution is often more suitable than the Gaussian distribution to fit the distributions of unchanged and changed classes in the LR map. We will consider using half-generalized Gaussian distribution to analyze the absolute value of the LR map.

Appendix

In this section, we give proof that the expectation of the unchanged area in the LR map is equal to zero under the assumption that the SAR images are interfered with by multiplicative noise. This conclusion leads to (12).

Let I_1 and I_2 represent pixels of noise-free images of $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$. Let $\boldsymbol{N}_1$ and $\boldsymbol{N}_2$ denote the multiplicative speckle noise of two multitemporal SAR images, $\boldsymbol{x}_1$ and $\boldsymbol{x}_1$ represent the random variables in the two multitemporal images. Then, the multiplicative statistical model is [21], [22]

$$\begin{aligned} \boldsymbol{x}_1 &= I_1 \boldsymbol{N}_1 \\ \boldsymbol{x}_2 &= I_2 \boldsymbol{N}_2. \end{aligned} \tag{41}$$

By dividing the two equations (41) and then taking the logarithm, you can get the following equation:

$$\boldsymbol{z} = \boldsymbol{I} + \boldsymbol{N} \tag{42}$$

where $\boldsymbol{I} = \log(I_2/I_1)$, $\boldsymbol{N} = \log(\boldsymbol{N}_2/\boldsymbol{N}_1)$, and $\boldsymbol{z} = \log(\boldsymbol{x}_2/\boldsymbol{x}_1)$.

If there is no change, then $\boldsymbol{I} = 0$. Then

$$E[\boldsymbol{z}] = E[\boldsymbol{I}] + E[\boldsymbol{N}] = E[\boldsymbol{N}]. \tag{43}$$

If it could be proved that $E[\boldsymbol{N}] = 0$, The proposition will be proved. It should be noticed that

$$E[\boldsymbol{N}] = E\big[\log(\boldsymbol{N}_2)\big] - E\big[\log(\boldsymbol{N}_1)\big]. \tag{44}$$

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Let $g(x) = \log(x)$ and $\mu = E[N_1] = E[N_2]$. The Taylor expansion of the expectation of the function of the random variable at μ [46] is defined as follows:

$$\begin{aligned} E\big[\log(N_1)\big] &= E[g(N_1)] \\ &= E\left[\sum_{k=0}^{\infty} \frac{g^{(k)}(\mu)}{k!}(N_1-\mu)^k\right] \\ &= \sum_{k=0}^{\infty} \frac{g^{(k)}(\mu)}{k!} E\big[(N_1-\mu)^k\big] \\ &= \sum_{k=0}^{\infty} \frac{g^{(k)}(\mu)}{k!}\mu_k \end{aligned} \tag{45}$$

where μ_k represents the central moment of order k of N_1. Since the speckle noise of the two images follows the same distribution, the moments of N_1 are the same as N_2, so

$$E[N] = 0. \tag{46}$$

Proposition proved.

In this section, we give proof that the expectation of the unchanged area in the ID is also equal to zero under the same noise model as (41).

Rewrite (41), and the multiplicative noise can be written in the form of additive noise

$$\begin{aligned} x_1 &= I_1 + I_1(N_1 - 1) = I_1 + V_1 \\ x_2 &= I_2 + I_2(N_2 - 1) = I_2 + V_2. \end{aligned} \tag{47}$$

According to [21], the expectation of N_1 and N_2 in (41) equal to 1

$$E[N_1] = E[N_2] = 1 \tag{48}$$

where V_1 and V_2 are random variables, the expectation of V_1 and V_2 are

$$\begin{aligned} E[V_1] &= E[I_1(N_1 - 1)] = 0 \\ E[V_2] &= E[I_2(N_2 - 1)] = 0. \end{aligned}$$

Subtracting the two equations in (47), then we get

$$x_2 - x_1 = I_2 - I_1 + V_2 - V_1.$$

Let $\mathcal{Z}_d$ represents $I_2 - I_1$ and z represents $x_2 - x_1$, then we have

$$z = \mathcal{Z}_d + V_2 - V_1. \tag{49}$$

Note that z denotes the ID. The aim is to prove that the expectation of z equals 0 under the condition of no change.

If there is no change, then $\mathcal{Z}_d = 0$, and we can get the expectation of z

$$E[z] = \mathcal{Z}_d + E[V_2 - V_1] = 0. \tag{50}$$

Proposition proved.

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